UNIT - 1

SIMPLE STRESSES & STRAINS

Mechanics of Solids (23A03302) - handwritten class notes

Topics covered in this unit

- Elasticity & plasticity, types of stresses & strains, Hooke's law
- Stress-strain diagram for mild steel, working stress, factor of safety
- Lateral strain, Poisson's ratio, volumetric strain
- Bars of varying section, composite bars, temperature stresses
- Complex stresses: inclined planes, principal planes & stresses, Mohr's circle
- Relation between elastic constants
- Strain energy, resilience: gradual, sudden, impact & shock loading

Tip: learn the formulas in the yellow boxes - most exam problems are direct substitution!

1. Introduction

Mechanics of solids (strength of materials) studies how a deformable body responds to loads - how much it deforms and whether it fails.

When a body is loaded, it develops an internal resisting force. This resisting force per unit area is called stress.

- Rigid body: shape does not change (ideal).
- Deformable body: changes shape and size under load (real).

- Aim of design: strength, stiffness and stability at minimum cost.

2. Elasticity and plasticity

Elasticity: the property by which a body regains its original size and shape when the load is removed.

Plasticity: the property by which a body retains permanent deformation after the load is removed.

- Perfectly elastic body – returns completely, e.g. steel within elastic limit, rubber.
- Plastic body – no recovery at all, e.g. wet clay, lead.
- Elastic limit: the maximum stress up to which the material is elastic.
- Ductile: large plastic deformation before fracture (mild steel, copper). Brittle: fractures with little strain (cast iron, glass, concrete).

3. Stress and its types

Stress σ = Load P / Area A unit: N/mm^2 = MPa

Type	Action of load	Formula
Tensile	pulls, bar elongates	$\sigma_t = P/A$
Compressive	pushes, bar shortens	$\sigma_c = P/A$
Shear	force parallel to the area	$\tau = P/A$
Bearing	contact pressure (rivets)	$\sigma_b = P/(d \cdot t)$

1 MPa = 1 N/mm^2 = 10^6 N/m^2; 1 GPa = 10^3 N/mm^2 = 10^9 Pa. E for steel = 2×10^5 N/mm^2.

4. Strain and its types

Strain is the change in dimension per unit original dimension. It has no units.

- Linear (tensile / compressive) strain $\varepsilon = \delta L / L$
- Shear strain $\varphi = \delta / h$ (angle of distortion in radians)
- Volumetric strain $\varepsilon_v = \delta V / V$

5. Hooke's law

Within the proportional limit, stress is directly proportional to the strain it produces.

$$\sigma \propto \varepsilon \quad \Rightarrow \quad \sigma = E\,\varepsilon \qquad E = \text{Young's modulus}$$
$$\tau = G\,\varphi \qquad K = \sigma / \varepsilon_v \text{ (bulk modulus)}$$

Combining $\sigma = P/A$ and $\varepsilon = \delta L/L$, the elongation of an axially loaded bar is:

$$\delta L = P L / (A E)$$

E, G and K are the three elastic moduli. A stiff material has a large E (steel 200 GPa, aluminium 70 GPa).

6. Stress-strain diagram for mild steel



Fig: Stress-strain curve of mild steel (schematic, not to scale)

- O-A: straight line, Hooke's law valid. A = proportional limit.
- A-B: elastic but not proportional. B = elastic limit.
- C, D: yielding. Upper yield point C, then stress drops to lower yield point D; strain increases with almost constant stress.

- D-E: strain hardening – the material stiffens, stress rises to the ultimate stress at E.
- E-F: necking – the area reduces locally and the bar breaks at F.

For materials with no clear yield point (aluminium, high-carbon steel) the 0.2 % proof stress (offset method) is used as yield stress.

> % elongation = $(L_f - L) / L \times 100$ % reduction in area = $(A - A_f) / A \times 100$

7. Working stress and factor of safety

The working (permissible) stress is the maximum stress allowed in service. It is kept below the failure stress to give a margin of safety.

> Factor of safety (FOS) = Ultimate stress / Working stress (for ductile materials, yield stress is used in place of ultimate stress)

Typical FOS: steel 3 to 4, cast iron 5 to 6, timber 7 to 10, dynamic loads larger.

Margin of safety = FOS – 1. Reasons for FOS: uncertain loads, material defects, workmanship, wear.

8. Lateral strain and Poisson's ratio

When a bar is stretched axially it also contracts laterally (and vice versa on compression).

> Poisson's ratio μ = lateral strain / longitudinal strain

- $\mu \approx 0.25$ to 0.33 for most metals (steel 0.3), concrete ≈ 0.15, rubber ≈ 0.5.
- Lateral strain is opposite in sign to the longitudinal strain: $\varepsilon_{lat} = -\mu \varepsilon$.

9. Volumetric strain

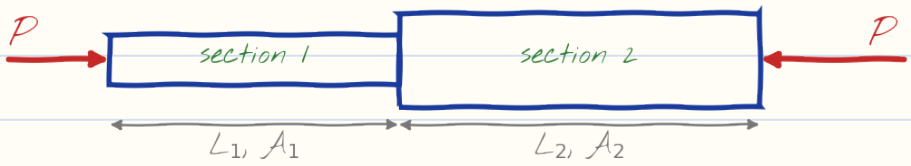
For a rectangular bar under axial stress σ along x:

$$\varepsilon_v = \varepsilon_x + \varepsilon_y + \varepsilon_z = \varepsilon\,(1 - 2\mu) \quad [\varepsilon = \sigma/E]$$

For a circular bar: $\varepsilon_v = \varepsilon_L + 2\,\varepsilon_d = \delta L/L + 2\,\delta d/d$. Since ε_v cannot be negative for tension, $\mu \le 0.5$.

10. Bars of varying section

A bar made of several lengths with different areas under the same axial load P (same material):



$$\delta L = \Sigma\, P\, L_i / (A_i\, E) = (P/E)\,[L_1/A_1 + L_2/A_2 + \ldots]$$

If loads act at the junctions, draw the axial-force diagram first and use the force of each segment.

- Tapered circular bar (d_1 to d_2): $\delta L = 4\,P\,L / (\pi\,E\,d_1\,d_2)$
- Tapered rectangular bar (constant thickness t, widths b_1, b_2): $\delta L = P\,L\,\ln(b_1/b_2) / [E\,t\,(b_1 - b_2)]$
- Bar hanging under own weight W: $\delta L = W\,L / (2\,A\,E) = \rho\,g\,L^2 / (2E)$

Example 1: stepped bar

A steel bar carries $P = 50$ KN. Section 1: $d = 20$ mm, $L = 500$ mm. Section 2: $d = 30$ mm, $L = 400$ mm. $E = 2\times10^5$ MPa. Find δL.

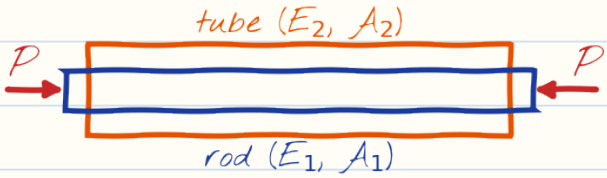
$A_1 = \pi/4 \times 20^2 = 314.16$ mm^2, $A_2 = \pi/4 \times 30^2 = 706.86$ mm^2

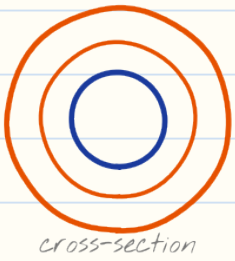
$\delta L = (50000 / 2\times10^5)\,[500/314.16 + 400/706.86]$

$= 0.25 \times [1.5915 + 0.5659] = 0.539$ mm

11. Composite bars

Two (or more) materials joined so that they extend or shorten together under a common load.





- Equilibrium: $P = P_1 + P_2 = \sigma_1 A_1 + \sigma_2 A_2$
- Compatibility: $\varepsilon_1 = \varepsilon_2 \Rightarrow \sigma_1/E_1 = \sigma_2/E_2$ (equal change in length)

$$P_1 = P \times A_1E_1 / (A_1E_1 + A_2E_2) \qquad \sigma_1/\sigma_2 = E_1/E_2 = \text{modular ratio } m$$

Example 2: steel rod in copper tube

Steel rod $d = 25$ mm inside a copper tube (40 mm outer, 30 mm inner) carries $P = 100$ KN compression. $E_s = 200$ GPa, $E_c = 100$ GPa.

$A_s = 490.87\ mm^2$, $A_c = \pi/4\,(40^2 - 30^2) = 549.78\ mm^2$

$\sigma_s = 2\,\sigma_c$ and $100000 = \sigma_c\,(2 \times 490.87 + 549.78) = 1531.5\,\sigma_c$

$\sigma_c = 65.3$ MPa, $\sigma_s = 130.6$ MPa

Check: $P_s = 64.1$ KN, $P_c = 35.9$ KN, sum = 100 KN ✓

12. Temperature stresses

A bar free to expand grows by $\delta = \alpha T L$ (no stress). If the expansion is prevented, a thermal stress develops:

$$\sigma = \alpha T E \qquad P = \alpha T E A$$

(α = coefficient of thermal expansion, T = temperature rise)

- Rise in temperature + fixed ends → compressive stress; fall in temperature → tensile stress.
- If the support yields by δ: $\sigma = E(\alpha T L - \delta)/L$.

- Composite bar heated: the material with larger α is put in compression, the other in tension.

$$\sigma_1 A_1 = \sigma_2 A_2 \text{ (equilibrium)} \qquad \sigma_1/E_1 + \sigma_2/E_2 = (\alpha_2 - \alpha_1) T \text{ (compatibility)}$$

Example 3

A steel rod of 25 mm dia is fixed between rigid walls at 20 °C and heated to 70 °C. $\alpha = 12\times10^{-6}$ /°C, $E = 200$ GPa.

$\sigma = \alpha T E = 12\times10^{-6} \times 50 \times 2\times10^{5} = 120$ MPa (compressive)

$P = \sigma A = 120 \times 490.87 = 58\,904\ N \approx 58.9$ KN

13. Complex stresses – stresses on an inclined plane

Case 1: uniaxial stress σ. On a plane whose normal makes angle θ with the axis:

$$\sigma_n = \sigma \cos^2\theta \qquad \tau_t = (\sigma/2) \sin 2\theta$$

Maximum shear stress = $\sigma/2$, occurring on a plane at 45° to the axis. Max normal stress = σ at $\theta = 0$.

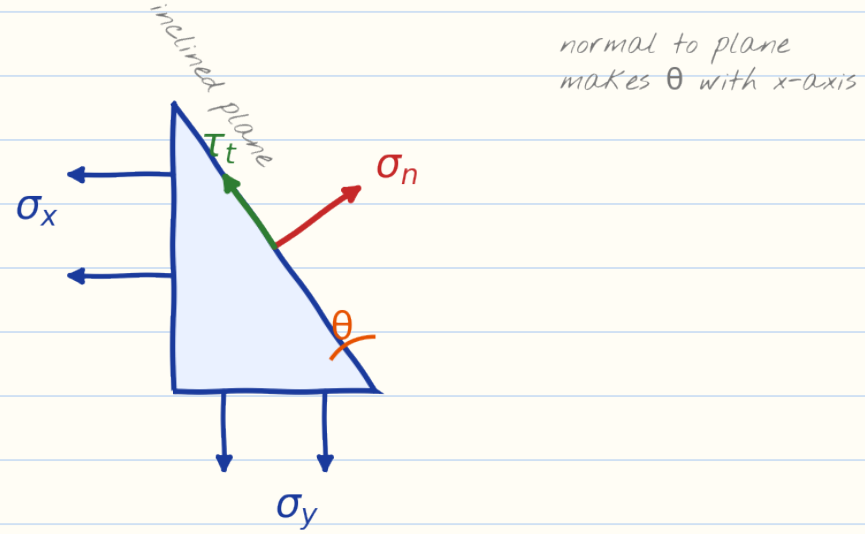


Fig: element cut by an inclined plane (normal at to x-axis)

Case 2: biaxial stresses σ_x, σ_y

$$\sigma_n = (\sigma_x+\sigma_y)/2 + (\sigma_x-\sigma_y)/2 \cdot \cos 2\theta$$
$$\tau_t = (\sigma_x-\sigma_y)/2 \cdot \sin 2\theta$$

Case 3: biaxial + shear τ_{xy}

$$\sigma_n = (\sigma_x+\sigma_y)/2 + (\sigma_x-\sigma_y)/2 \cdot \cos 2\theta + \tau_{xy} \sin 2\theta$$
$$\tau_t = (\sigma_x-\sigma_y)/2 \cdot \sin 2\theta - \tau_{xy} \cos 2\theta$$

- Resultant stress on the plane $\sigma_R = \sqrt{(\sigma_n^2 + \tau_t^2)}$
- Obliquity (angle of resultant with the normal): $\tan \varphi = \tau_t / \sigma_n$

14. Principal planes and principal stresses

Principal planes are planes on which the shear stress is zero. The normal stresses on them are the principal stresses.

$$\tan 2\theta_p = 2\tau_{xy} / (\sigma_x - \sigma_y)$$
$$\sigma_{1,2} = (\sigma_x+\sigma_y)/2 \pm \sqrt{[((\sigma_x-\sigma_y)/2)^2 + \tau_{xy}^2]}$$
$$\tau_{max} = \sqrt{[((\sigma_x-\sigma_y)/2)^2 + \tau_{xy}^2]} = (\sigma_1 - \sigma_2)/2$$

- The two principal planes are 90° apart, the planes of max shear are at 45° to them.
- On planes of max shear the normal stress is $(\sigma_x+\sigma_y)/2$.

Example 4

σ_x = 100 MPa, σ_y = 40 MPa (both tensile), τ_{xy} = 30 MPa. Find principal stresses and the stresses on a plane at θ = 30°.

average = 70, $R = \sqrt{(30^2 + 30^2)} = 42.43$

σ_1 = 112.43 MPa, σ_2 = 27.57 MPa, τ_{max} = 42.43 MPa

$\tan 2\theta_p = 60/60 = 1 \Rightarrow 2\theta_p = 45°$, $\theta_p = 22.5°$

at 30°: $\sigma_n = 70 + 30 \cos 60° + 30 \sin 60° = 110.98$ MPa

$\tau_t = 30 \sin 60° - 30 \cos 60° = 10.98$ MPa

15. Mohr's circle

A graphical method - every point on the circle represents the (σ, τ) on some plane. Angles on the circle are double the real angles.

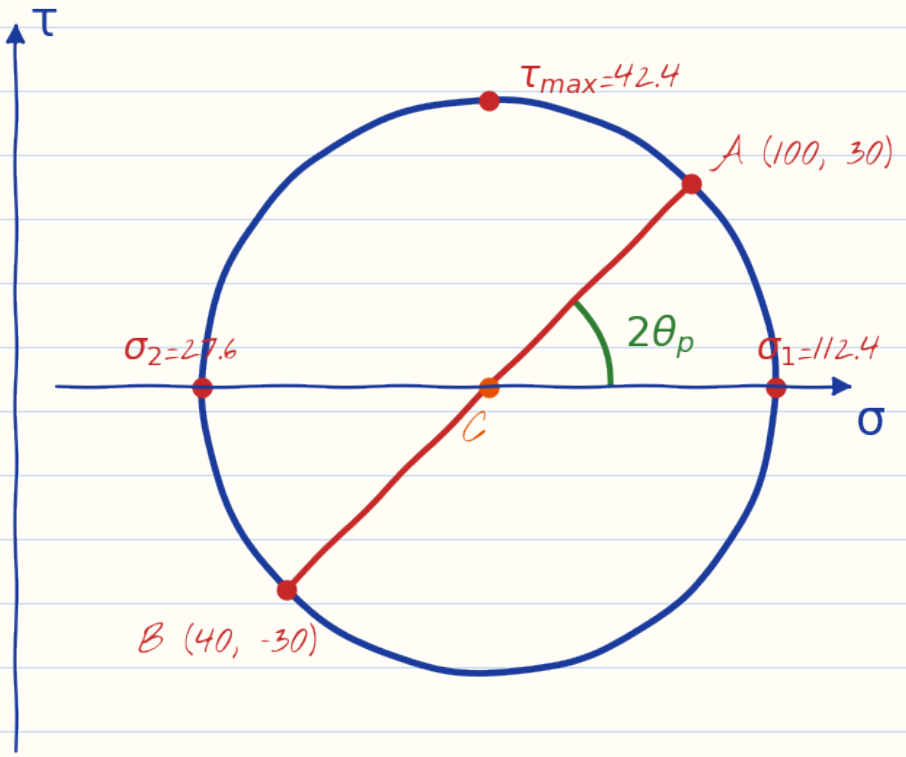


Fig: Mohr's circle for Example 4

- Step 1: choose scale, plot A (σ_x, $+\tau_{xy}$) and B (σ_y, $-\tau_{xy}$).
- Step 2: join AB, it cuts the σ-axis at the centre C = $(\sigma_x+\sigma_y)/2$.
- Step 3: draw the circle with diameter AB. Radius = τ_{max}.
- Step 4: the points where the circle cuts the σ-axis give σ_1 and σ_2.
- Step 5: for a plane at θ, rotate CA by 2θ (same sense), the coordinates of that point are σ_n, τ_t.

16. Relation between elastic constants

$E = 2G(1+\mu)$	$E = 3K(1-2\mu)$
$E = 9KG/(3K+G)$	$\mu = (3K-2G)/(6K+2G)$

Proof of $E = 2G(1+\mu)$: in pure shear τ the principal stresses are $+\tau$ and $-\tau$ at 45°. Strain along $+\tau$ direction: $\varepsilon = \tau/E + \mu\tau/E = (\tau/E)(1+\mu)$. This equals half the shear strain: $\varphi/2 = \tau/2G$. Hence $E = 2G(1+\mu)$.

Proof of $E = 3K(1-2\mu)$: under equal stress σ on all faces $\varepsilon_v = 3\sigma/E(1-2\mu)$ and $\varepsilon_v = \sigma/K$. Hence $E = 3K(1-2\mu)$.

Example 5

$E = 2\times10^5$ MPa, $G = 0.8\times10^5$ MPa. Find μ and K.

$\mu = E/2G - 1 = 2/1.6 - 1 = 0.25$

$K = E / 3(1 - 2\mu) = 2\times10^5 / 1.5 = 1.333\times10^5$ MPa

17. Strain energy and resilience

Work stored in a deformed elastic body = strain energy U. For gradual loading, $U = \frac{1}{2} P \delta L$ (area under the load-extension line).

$$U = \sigma^2 V / (2E) \qquad [V = A L = \text{volume}]$$

- Proof resilience: maximum strain energy stored (at elastic limit).
- Modulus of resilience = $\sigma_p^2 / 2E$ (energy per unit volume).
- Shear strain energy: $U = \tau^2 V / (2G)$.

18. Gradual, sudden, impact and shock loading

Loading	Stress	Elongation
Gradual (static)	$\sigma = P/A$	$\delta = PL/AE$
Sudden (suddenly applied)	$\sigma = 2P/A$	$\delta = 2PL/AE$
Impact (fall through h)	$\sigma = (P/A)[1 + \sqrt{(1 + 2hAE/PL)}]$	$\delta = \sigma L/E$

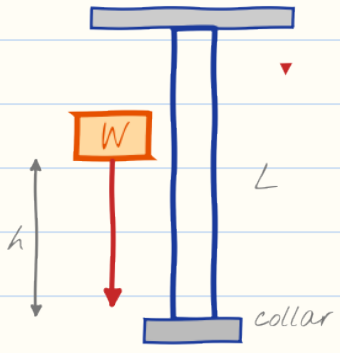


Fig: weight W falling through height h on a collar

- Sudden load produces twice the stress of the same load applied gradually.
- If h is large compared with δ: $\sigma \approx \sqrt{(2 W h E / A L)}$.
- Shock load: a sudden/impact load whose effect is judged by the energy absorbed.

Example 6: impact load

$W = 2$ KN falls $h = 30$ mm on a collar at the end of a 3 m long, 20 mm dia bar. $E = 200$ GPa.

$A = 314.16$ mm^2, $W/A = 6.366$ MPa, $\delta_{st} = WL/AE = 0.0955$ mm

$\sigma = 6.366\,[1 + \sqrt{(1 + 2 \times 30/0.0955)}] = 6.366 \times 26.09 = 166$ MPa

More solved problems

Example 7: factor of safety

A mild steel bar (yield stress 250 MPa) is to carry a tensile load with FOS = 2.5. Find the safe load for $d = 20$ mm.

$\sigma_w = 250 / 2.5 = 100$ MPa, $A = 314.16$ mm^2

$P = \sigma_w A = 100 \times 314.16 = 31.4$ KN

Example 8: Poisson's ratio and volume change

A 20 mm dia, 2 m long bar carries 40 KN tension. $E = 2 \times 10^5$ MPa, $\mu = 0.3$. Find δL, δd and δV.

$\sigma = 40000/314.16 = 127.3$ MPa, $\varepsilon = \sigma/E = 6.366 \times 10^{-4}$

$\delta L = \varepsilon L = 6.366 \times 10^{-4} \times 2000 = 1.273$ mm

$\delta d = -\mu \varepsilon d = -0.3 \times 6.366 \times 10^{-4} \times 20 = -0.0038$ mm

$\varepsilon_v = \varepsilon (1 - 2\mu) = 6.366 \times 10^{-4} \times 0.4 = 2.546 \times 10^{-4}$

$V = 314.16 \times 2000 = 6.283 \times 10^5$ mm^3 $\Rightarrow$ $\delta V = 160$ mm^3

Example 9: composite bar heated

Steel rod ($A = 600$ mm^2, $E = 200$ GPa, $\alpha = 12 \times 10^{-6}$) and copper rod ($A = 900$ mm^2, $E = 100$ GPa, $\alpha = 17 \times 10^{-6}$) are rigidly joined at both ends and heated by 60 °C. Find stresses.

Equilibrium: $\sigma_s \times 600 = \sigma_c \times 900$ $\Rightarrow$ $\sigma_s = 1.5\,\sigma_c$

Compatibility: $\sigma_s/E_s + \sigma_c/E_c = (\alpha_c - \alpha_s)\,T = 5\times10^{-6} \times 60 = 3\times10^{-4}$

$1.5\sigma_c/2\times10^5 + \sigma_c/10^5 = 3\times10^{-4} \Rightarrow 3.5\,\sigma_c = 60$

$\sigma_c = 17.14$ MPa (compressive), $\sigma_s = 25.71$ MPa (tensile)

Check forces: $25.71 \times 600 = 17.14 \times 900 = 15.43$ KN ✓

Example 10: inclined plane, uniaxial

A bar carries a uniaxial tensile stress of 100 MPa. Find σ_n and τ on a plane whose normal is at 30° to the axis.

$\sigma_n = 100\cos^2 30° = 75$ MPa

$\tau = (100/2)\sin 60° = 43.3$ MPa (maximum shear = 50 MPa at 45°)

Example 11: strain energy

A bar 1 m long, $A = 500$ mm², carries 50 KN gradually. $E = 200$ GPa. Find strain energy stored.

$\sigma = 100$ MPa, $V = 500 \times 1000 = 5\times10^5$ mm³

$U = \sigma^2 V / 2E = 100^2 \times 5\times10^5 / (2 \times 2\times10^5) = 12\,500$ N·mm = 12.5 N·m (J)

If the same load is applied suddenly: $\sigma = 200$ MPa, $U = 4 \times 12.5 = 50$ J.

Important questions (university pattern)

- Draw the stress-strain curve for mild steel and explain salient points.
- Define Poisson's ratio and derive the relation between E, G and K.
- Derive the expression for stress on an inclined plane under biaxial stress with shear.
- Explain Mohr's circle and use it to find principal stresses for a given state of stress.
- Derive the elongation of a bar of uniformly tapering circular section.

- A composite bar is heated - derive stresses in the two materials.
- Derive the expression for stress due to a falling weight (impact load).

Quick revision - Unit I

$\sigma = P/A$	$\varepsilon = \delta L/L$	$\delta L = PL/AE$	$FOS = \sigma_{"}/\sigma_{""}$	μ = lateral/long.
$\sigma_{temp} = \alpha TE$	$\sigma_{1,2} = (\sigma_x + \sigma_y)/2 \pm \sqrt{[\ldots]}$	$U = \sigma^2 V/2E$		

- Important: stress-strain curve of mild steel; relation between E, G, K; Mohr's circle problems.
- Important: composite bar & temperature stress problems; principal stress problems; impact loading.